

Sequences and Series of Real Numbers (2022–2026)

Category: Real Analysis • Official Mock Test Series

TOTAL QUESTIONS

49

TOTAL MARKS

81 M

TEST DURATION

147 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2022–20)

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	30	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	7	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	12	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q43 • ID: JAM_2026_Q43)

NAT

+1 M

Neg: Nil (0)

Q.43

$$\lim_{n \rightarrow \infty} \left(\frac{n^2+1}{\sqrt{n^6+1}} + \cdots + \frac{n^2+n}{\sqrt{n^6+n}} \right) = \text{_____} (\text{rounded off to one decimal place}).$$

Question 2 (JAM 2026 • Q42 • ID: JAM_2026_Q42)

NAT

+1 M

Neg: Nil (0)

Q.42

$$\int_0^1 \left(\sum_{k=1}^{\infty} \frac{(\log_e 2)^k x^{k-1}}{(k-1)!} \right) dx = \text{_____} (\text{rounded off to one decimal place}).$$

Question 3 (JAM 2026 • Q41 • ID: JAM_2026_Q41)

NAT

+1 M

Neg: Nil (0)

Q.41

The radius of convergence of the series $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!} (\log_e n)^{-1} x^n$ is _____
(rounded off to one decimal place).

Question 4 (JAM 2026 • Q32 • ID: JAM_2026_Q32)

MSQ

+2 M

Neg: Nil (0)

Q.32	Which of the following statements is/are TRUE?
(A)	There exists a monotone sequence that does not converge but has a convergent subsequence.
(B)	There exists a sequence that has a bounded subsequence but does not have any convergent subsequence.
(C)	There exists a sequence (x_n) such that given any positive integer m , (x_n) has a subsequence converging to m .
(D)	There exists a sequence (x_n) such that $(x_{n+1} - x_n)$ converges to 0 but (x_n) does not converge.

Question 5 (JAM 2026 • Q29 • ID: JAM_2026_Q29)

MCQ

+2 M

Neg: -0.67

Q.29

For each positive integer n , let

$$x_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n} - \log_e n$$

and

$$y_n = \int_1^n \frac{\cos t}{t^2} dt .$$

Which ONE of the following statements about the sequences (x_n) and (y_n) is TRUE?(A) (x_n) and (y_n) are convergent.(B) (x_n) is convergent and (y_n) is not convergent.(C) (x_n) is not convergent and (y_n) is convergent.(D) (x_n) is not convergent and (y_n) is not convergent.

Question 6 (JAM 2026 • Q12 • ID: JAM_2026_Q12)

MCQ

+2 M

Neg: -0.67

Q.12	For which one of the following pairs of values of a and b , does the series $\sum_{k=1}^{\infty} \frac{(3x)^k}{2^{\sqrt{k}}(1-x)^k}$ converge for all $x \in (a, b)$?
(A)	$a = \frac{-1}{4}$ and $b = \frac{1}{2}$
(B)	$a = -1$ and $b = \frac{1}{5}$
(C)	$a = \frac{-1}{2}$ and $b = \frac{1}{4}$
(D)	$a = \frac{-1}{4}$ and $b = \frac{1}{3}$

Question 7 (JAM 2026 • Q11 • ID: JAM_2026_Q11)

MCQ

+2 M

Neg: -0.67

Q.11	Given a sequence of real numbers (a_n), define $b_n = \begin{cases} a_n, & \text{if } a_n \geq 0 \\ 0, & \text{if } a_n < 0 \end{cases} \text{ and } c_n = \begin{cases} a_n, & \text{if } a_n < 0 \\ 0, & \text{if } a_n \geq 0 \end{cases}$ for each positive integer n. Which ONE of the following statements is TRUE?
(A)	If $\sum_{n=1}^{\infty} a_n$ does not converge, then $\sum_{n=1}^{\infty} b_n$ does not converge.
(B)	If $\sum_{n=1}^{\infty} a_n$ converges but $\sum_{n=1}^{\infty} a_n $ does not converge, then $\sum_{n=1}^{\infty} b_n$ converges.
(C)	If $\sum_{n=1}^{\infty} a_n$ converges but $\sum_{n=1}^{\infty} a_n $ does not converge, then $\sum_{n=1}^{\infty} b_n$ does not converge.
(D)	If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} b_n$ converges or $\sum_{n=1}^{\infty} c_n$ converges.

Question 8 (JAM 2026 • Q2 • ID: JAM_2026_Q2)

MCQ

+1 M

Neg: -0.33

Q.2	For each positive integer n , define $x_n = (-1)^n$. Which ONE of the following statements about the sequence (x_n) is FALSE?
(A)	There exists $\epsilon > 0$ such that $ x_n - 1 < \epsilon$ for all positive integers n .
(B)	There exists $\epsilon > 0$ such that for all $M > 0$ there exists a positive integer $N > M$ for which $ x_N - 1 > \epsilon$.
(C)	For all $\epsilon > 0$ and $M > 0$ there exists a positive integer N such that $N > M$ and $ x_N - 1 < \epsilon$.
(D)	For all $\epsilon > 0$ and $M > 0$ there exists a positive integer N such that $N > M$ and $ x_N - 1 > \epsilon$.

Question 9 (JAM 2026 • Q1 • ID: JAM_2026_Q1)

MCQ

+1 M

Neg: -0.33

Q.1	For each positive integer n , let $x_n = 1 - (-1)^n + \frac{1}{n}$ and $y_n = \left(1 + \frac{1}{2n}\right)^{3n}$. Which ONE of the following statements about the sequences (x_n) and (y_n) is TRUE?
(A)	(x_n) and (y_n) are convergent.
(B)	(x_n) is not convergent and (y_n) is not convergent.
(C)	(x_n) is convergent and (y_n) is not convergent.
(D)	(x_n) is not convergent and (y_n) is convergent.

Question 10 (JAM 2025 • Q54 • ID: JAM_2025_Q54)

NAT

+2 M

Neg: Nil (0)

Q.54 The value of the infinite series

$$\sum_{n=1}^{\infty} n \left(\frac{3}{4}\right)^{2(n-1)}$$

is equal to _____ (rounded off to two decimal places)

Question 11 (JAM 2025 • Q41 • ID: JAM_2025_Q41)

NAT

+1 M

Neg: Nil (0)

Q.41 The radius of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{\left(x + \frac{1}{4}\right)^n}{(-2)^n n^2}$$

about $x = -\frac{1}{4}$, is equal to _____ (rounded off to two decimal places).

Question 12 (JAM 2025 • Q36 • ID: JAM_2025_Q36)

MSQ

+2 M

Neg: Nil (0)

Q.36 For $n \in \mathbb{N}$, let

$$x_n = \sum_{k=1}^n \frac{k}{n^2 + k}.$$

Then, which of the following is/are TRUE?

- (A) The sequence (x_n) converges
- (B) The series $\sum_{n=1}^{\infty} x_n$ converges
- (C) The series $\sum_{n=1}^{\infty} x_n$ does NOT converge
- (D) The series $\sum_{n=1}^{\infty} x_n^n$ converges

Question 13 (JAM 2025 • Q26 • ID: JAM_2025_Q26)

MCQ

+2 M

Neg: -0.67

Q.26 Let $x_1 > 0$. For $n \in \mathbb{N}$, define $x_{n+1} = x_n + 4$. If

$$\lim_{n \rightarrow \infty} \left(\frac{1}{x_2 x_3} + \frac{1}{x_3 x_4} + \cdots + \frac{1}{x_{n+1} x_{n+2}} \right) = \frac{1}{24},$$

then the value of x_1 is equal to

- (A) 1
- (B) 2
- (C) 3
- (D) 8

Question 14 (JAM 2025 • Q25 • ID: JAM_2025_Q25)

MCQ

+2 M

Neg: -0.67

Q.25 Let $x_1 = 1$. For $n \in \mathbb{N}$, define

$$x_{n+1} = \left(\frac{1}{2} + \frac{\sin^2 n}{n} \right) x_n.$$

Then, which one of the following is TRUE?

- (A) $\sum_{n=1}^{\infty} x_n$ converges
- (B) $\sum_{n=1}^{\infty} x_n$ does NOT converge
- (C) $\sum_{n=1}^{\infty} x_n^2$ does NOT converge
- (D) $\sum_{n=1}^{\infty} x_n x_{n+1}$ does NOT converge

Question 15 (JAM 2025 • Q21 • ID: JAM_2025_Q21)

MCQ

+2 M

Neg: -0.67

Q.21 For $n \in \mathbb{N}$, define x_n and y_n by

$$x_n = (-1)^n \frac{3^n}{n^3} \text{ and } y_n = (4^n + (-1)^n 3^n)^{1/n}.$$

Then, which one of the following is TRUE?

- (A) (x_n) has a convergent subsequence, and NO subsequence of (y_n) is convergent
- (B) NO subsequence of (x_n) is convergent, and (y_n) has a convergent subsequence
- (C) (x_n) has a convergent subsequence, and (y_n) has a convergent subsequence
- (D) NO subsequence of (x_n) is convergent, and NO subsequence of (y_n) is convergent

Question 16 (JAM 2025 • Q20 • ID: JAM_2025_Q20)

MCQ

+2 M

Neg: -0.67

Q.20

Let $x_1 = 2$ and $x_{n+1} = 2 + \frac{1}{2x_n}$ for all $n \in \mathbb{N}$.

Then, which one of the following is TRUE?

- (A) $x_{n+1} \geq \frac{4}{x_n}$ for all $n \in \mathbb{N}$, and (x_n) is a Cauchy sequence
- (B) $x_{n+1} < \frac{4}{x_n}$ for some $n \in \mathbb{N}$, and (x_n) is a Cauchy sequence
- (C) $x_{n+1} \geq \frac{4}{x_n}$ for all $n \in \mathbb{N}$, and (x_n) is NOT a Cauchy sequence
- (D) $x_{n+1} < \frac{4}{x_n}$ for some $n \in \mathbb{N}$, and (x_n) is NOT a Cauchy sequence

Question 17 (JAM 2025 • Q19 • ID: JAM_2025_Q19)

MCQ

+2 M

Neg: -0.67

Q.19

Let $x_1 = \frac{5}{2}$. For $n \in \mathbb{N}$, define

$$x_{n+1} = \frac{1}{5}(x_n^2 + 6).$$

Then, which one of the following is TRUE?

- (A) (x_n) is an increasing sequence, and (x_n) is NOT a bounded sequence
- (B) (x_n) is NOT an increasing sequence, and (x_n) is NOT a bounded sequence
- (C) (x_n) is NOT a decreasing sequence, and (x_n) is a bounded sequence
- (D) (x_n) is a decreasing sequence, and (x_n) is a bounded sequence

Question 18 (JAM 2025 • Q18 • ID: JAM_2025_Q18)

MCQ

+2 M

Neg: -0.67

Q.18 For $n \in \mathbb{N}$, define x_n and y_n by

$$x_n = (-1)^n \cos \frac{1}{n} \quad \text{and} \quad y_n = \sum_{k=1}^n \frac{1}{n+k}.$$

Then, which one of the following is TRUE?

- (A) $\sum_{n=1}^{\infty} x_n$ converges, and $\sum_{n=1}^{\infty} y_n$ does NOT converge
- (B) $\sum_{n=1}^{\infty} x_n$ does NOT converge, and $\sum_{n=1}^{\infty} y_n$ converges
- (C) $\sum_{n=1}^{\infty} x_n$ converges, and $\sum_{n=1}^{\infty} y_n$ converges
- (D) $\sum_{n=1}^{\infty} x_n$ does NOT converge, and $\sum_{n=1}^{\infty} y_n$ does NOT converge

Question 19 (JAM 2025 • Q1 • ID: JAM_2025_Q1)

MCQ

+1 M

Neg: -0.33

Q.1 The sum of the infinite series

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\pi^{2n+1}}{2^{2n+1} (2n)!}$$

is equal to

(A) $-\pi$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$

(D) $-\frac{\pi}{4}$

Question 20 (JAM 2024 • Q59 • ID: JAM_2024_Q59)

NAT

+2 M

Neg: Nil (0)

Q.59 For $n \in \mathbb{N}$, let

$$a_n = \frac{1}{n^{n-1}} \sum_{k=0}^n \frac{n!}{k! (n-k)!} \frac{n^k}{k+1}$$

and $\beta = \lim_{n \rightarrow \infty} a_n$. Then, the value of $\log \beta$ equals _____ (rounded off to two decimal places).

Question 21 (JAM 2024 • Q55 • ID: JAM_2024_Q55)

NAT

+2 M

Neg: Nil (0)

Q.55 For $n \in \mathbb{N}$, if

$$a_n = \frac{1}{n^3 + 1} + \frac{2^2}{n^3 + 2} + \cdots + \frac{n^2}{n^3 + n}$$

then the sequence $\{a_n\}_{n=1}^{\infty}$ converges to _____ (rounded off to two decimal places)

Question 22 (JAM 2024 • Q47 • ID: JAM_2024_Q47)

NAT

+1 M

Neg: Nil (0)

Q.47 Let $a_1 = 1, b_1 = 2$ and $c_1 = 3$. Consider the convergent sequences

$$\{a_n\}_{n=1}^{\infty}, \{b_n\}_{n=1}^{\infty} \text{ and } \{c_n\}_{n=1}^{\infty}$$

defined as follows:

$$a_{n+1} = \frac{a_n + b_n}{2}, \quad b_{n+1} = \frac{b_n + c_n}{2} \text{ and } c_{n+1} = \frac{c_n + a_n}{2} \text{ for } n \geq 1.$$

Then,

$$\sum_{n=1}^{\infty} b_n c_n (a_{n+1} - a_n) + \sum_{n=1}^{\infty} (b_{n+1} c_{n+1} - b_n c_n) a_{n+1}$$

equals _____ (rounded off to two decimal places)

Question 23 (JAM 2024 • Q39 • ID: JAM_2024_Q39)

MSQ

+2 M

Neg: Nil (0)

Q.39 For $0 < \alpha < 4$, define the sequence $\{x_n\}_{n=1}^{\infty}$ of real numbers as follows:

$$x_1 = \alpha \text{ and } x_{n+1} + 2 = -x_n(x_n - 4) \text{ for } n \in \mathbb{N}.$$

Which of the following is/are TRUE?

- (A) $\{x_n\}_{n=1}^{\infty}$ converges for at least three distinct values of $\alpha \in (0,1)$
- (B) $\{x_n\}_{n=1}^{\infty}$ converges for at least three distinct values of $\alpha \in (1,2)$
- (C) $\{x_n\}_{n=1}^{\infty}$ converges for at least three distinct values of $\alpha \in (2,3)$
- (D) $\{x_n\}_{n=1}^{\infty}$ converges for at least three distinct values of $\alpha \in (3,4)$

Question 24 (JAM 2024 • Q32 • ID: JAM_2024_Q32)

MSQ

+2 M

Neg: Nil (0)

Q.32 Which of the following statements is/are TRUE?

- (A) $\sum_{n=1}^{\infty} n \log \left(1 + \frac{1}{n^3}\right)$ is convergent
- (B) $\sum_{n=1}^{\infty} \left(1 - \cos \left(\frac{1}{n}\right)\right) \log n$ is convergent
- (C) $\sum_{n=1}^{\infty} n^2 \log \left(1 + \frac{1}{n^3}\right)$ is convergent
- (D) $\sum_{n=1}^{\infty} \left(1 - \cos \left(\frac{1}{\sqrt{n}}\right)\right) \log n$ is convergent

Question 25 (JAM 2024 • Q31 • ID: JAM_2024_Q31)

MSQ

+2 M

Neg: Nil (0)

Q.31 Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of real numbers.

Then, which of the following statements is/are always TRUE?

- (A) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_n^2$ converges absolutely
- (B) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_n^3$ converges absolutely
- (C) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_n^2$ converges
- (D) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_n^3$ converges

Question 26 (JAM 2024 • Q28 • ID: JAM_2024_Q28)

MCQ

+2 M

Neg: -0.67

Q.28 For $n \geq 3$, let a regular n -sided polygon P_n be circumscribed by a circle of radius R_n and let r_n be the radius of the circle inscribed in P_n . Then

$$\lim_{n \rightarrow \infty} \left(\frac{R_n}{r_n} \right)^{n^2}$$

equals

(A) $e^{(\pi^2)}$

(B) $e^{\left(\frac{\pi^2}{2}\right)}$

(C) $e^{\left(\frac{\pi^2}{3}\right)}$

(D) $e^{(2\pi^2)}$

Question 27 (JAM 2024 • Q21 • ID: JAM_2024_Q21)

MCQ

+2 M

Neg: -0.67

Q.21 For $p, q, r \in \mathbb{R}$, $r \neq 0$ and $n \in \mathbb{N}$, let

$$a_n = p^n n^q \left(\frac{n}{n+2} \right)^{n^2} \text{ and } b_n = \frac{n^n}{n! r^n} \left(\sqrt{\frac{n+2}{n}} \right).$$

Then, which one of the following statements is TRUE?

(A) If $1 < p < e^2$ and $q > 1$, then $\sum_{n=1}^{\infty} a_n$ is convergent

(B) If $e^2 < p < e^4$ and $q > 1$, then $\sum_{n=1}^{\infty} a_n$ is convergent

(C) If $1 < r < e$, then $\sum_{n=1}^{\infty} b_n$ is convergent

(D) If $\frac{1}{e} < r < 1$, then $\sum_{n=1}^{\infty} b_n$ is convergent

Question 28 (JAM 2024 • Q20 • ID: JAM_2024_Q20)

MCQ

+2 M

Neg: -0.67

Q.20 Define the sequences $\{a_n\}_{n=3}^{\infty}$ and $\{b_n\}_{n=3}^{\infty}$ as

$$a_n = (\log n + \log \log n)^{\log n} \quad \text{and} \quad b_n = n^{\left(1 + \frac{1}{\log n}\right)}.$$

Which one of the following is TRUE?

- (A) $\sum_{n=3}^{\infty} \frac{1}{a_n}$ is convergent but $\sum_{n=3}^{\infty} \frac{1}{b_n}$ is divergent
- (B) $\sum_{n=3}^{\infty} \frac{1}{a_n}$ is divergent but $\sum_{n=3}^{\infty} \frac{1}{b_n}$ is convergent
- (C) Both $\sum_{n=3}^{\infty} \frac{1}{a_n}$ and $\sum_{n=3}^{\infty} \frac{1}{b_n}$ are divergent
- (D) Both $\sum_{n=3}^{\infty} \frac{1}{a_n}$ and $\sum_{n=3}^{\infty} \frac{1}{b_n}$ are convergent

Question 29 (JAM 2024 • Q14 • ID: JAM_2024_Q14)

MCQ

+2 M

Neg: -0.67

Q.14 For $n \in \mathbb{N}$, let

$$a_n = \frac{1}{(3n+2)(3n+4)} \quad \text{and} \quad b_n = \frac{n^3 + \cos(3^n)}{3^n + n^3}.$$

Then, which one of the following is TRUE?

- (A) $\sum_{n=1}^{\infty} a_n$ is convergent but $\sum_{n=1}^{\infty} b_n$ is divergent
- (B) $\sum_{n=1}^{\infty} a_n$ is divergent but $\sum_{n=1}^{\infty} b_n$ is convergent
- (C) Both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are divergent
- (D) Both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent

Question 30 (JAM 2023 • Q50 • ID: JAM_2023_Q50)

NAT

+1 M

Neg: Nil (0)

Q. 50 The value of $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{2^n} + \frac{1}{3^n} + \cdots + \frac{1}{(2023)^n} \right)^{\frac{1}{n}}$ is equal to _____.

(rounded off to two decimal places)

Question 31 (JAM 2023 • Q49 • ID: JAM_2023_Q49)

NAT

+1 M

Neg: Nil (0)

Q. 49 The sum of the series $\sum_{n=1}^{\infty} \frac{2n+1}{(n^2+1)(n^2+2n+2)}$ is equal to _____.

(rounded off to two decimal places)

Question 32 (JAM 2023 • Q40 • ID: JAM_2023_Q40)

MSQ

+2 M

Neg: Nil (0)

Q. 40 Let R_1 and R_2 be the radii of convergence of the power series $\sum_{n=1}^{\infty} (-1)^n x^{n-1}$ and

$\sum_{n=1}^{\infty} (-1)^n \frac{x^{n+1}}{n(n+1)}$, respectively. Then

- (A) $R_1 = R_2$
- (B) $R_2 > 1$
- (C) $\sum_{n=1}^{\infty} (-1)^n x^{n-1}$ converges for all $x \in [-1, 1]$
- (D) $\sum_{n=1}^{\infty} (-1)^n \frac{x^{n+1}}{n(n+1)}$ converges for all $x \in [-1, 1]$

Question 33 (JAM 2023 • Q24 • ID: JAM_2023_Q24)

MCQ

+2 M

Neg: -0.67

Q. 24 Suppose $f : (-1, 1) \rightarrow \mathbb{R}$ is an infinitely differentiable function such that the series $\sum_{j=0}^{\infty} a_j \frac{x^j}{j!}$ converges to $f(x)$ for each $x \in (-1, 1)$, where,

$$a_j = \int_0^{\pi/2} \theta^j \cos^j(\tan \theta) d\theta + \int_{\pi/2}^{\pi} (\theta - \pi)^j \cos^j(\tan \theta) d\theta$$

for $j \geq 0$. Then

- (A) $f(x) = 0$ for all $x \in (-1, 1)$
- (B) f is a non-constant even function on $(-1, 1)$
- (C) f is a non-constant odd function on $(-1, 1)$
- (D) f is NEITHER an odd function NOR an even function on $(-1, 1)$

Question 34 (JAM 2023 • Q21 • ID: JAM_2023_Q21)

MCQ

+2 M

Neg: -0.67

Q. 21 Let (a_n) and (b_n) be sequences of real numbers such that

$$|a_n - a_{n+1}| = \frac{1}{2^n} \quad \text{and} \quad |b_n - b_{n+1}| = \frac{1}{\sqrt{n}} \quad \text{for } n \in \mathbb{N}.$$

Then

- (A) both (a_n) and (b_n) are Cauchy sequences
- (B) (a_n) is a Cauchy sequence but (b_n) need NOT be a Cauchy sequence
- (C) (a_n) need NOT be a Cauchy sequence but (b_n) is a Cauchy sequence
- (D) both (a_n) and (b_n) need NOT be Cauchy sequences

Question 35 (JAM 2023 • Q20 • ID: JAM_2023_Q20)

MCQ

+2 M

Neg: -0.67

Q. 20 Let (a_n) be a sequence of real numbers such that the series $\sum_{n=0}^{\infty} a_n(x-2)^n$ converges at $x = -5$. Then this series also converges at

- (A) $x = 9$
- (B) $x = 12$
- (C) $x = 5$
- (D) $x = -6$

Question 36 (JAM 2023 • Q19 • ID: JAM_2023_Q19)

MCQ

+2 M

Neg: -0.67

Q. 19 Let $a_n = \frac{1 + 2^{-2} + \cdots + n^{-2}}{n}$ for $n \in \mathbb{N}$. Then

- (A) both the sequence (a_n) and the series $\sum_{n=1}^{\infty} a_n$ are convergent
- (B) the sequence (a_n) is convergent but the series $\sum_{n=1}^{\infty} a_n$ is NOT convergent
- (C) both the sequence (a_n) and the series $\sum_{n=1}^{\infty} a_n$ are NOT convergent
- (D) the sequence (a_n) is NOT convergent but the series $\sum_{n=1}^{\infty} a_n$ is convergent

Question 37 (JAM 2023 • Q11 • ID: JAM_2023_Q11)

MCQ

+2 M

Neg: -0.67

Q. 11 Let $a_n = \sin\left(\frac{1}{n^3}\right)$ and $b_n = \sin\left(\frac{1}{n}\right)$ for $n \in \mathbb{N}$. Then

- (A) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent
- (B) $\sum_{n=1}^{\infty} a_n$ is convergent but $\sum_{n=1}^{\infty} b_n$ is NOT convergent
- (C) $\sum_{n=1}^{\infty} a_n$ is NOT convergent but $\sum_{n=1}^{\infty} b_n$ is convergent
- (D) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are NOT convergent

Question 38 (JAM 2023 • Q10 • ID: JAM_2023_Q10)

MCQ

+1 M

Neg: -0.33

Q. 10 Let (a_n) be a sequence of real numbers defined by

$$a_n = \begin{cases} 1 & \text{if } n \text{ is prime} \\ -1 & \text{if } n \text{ is not prime.} \end{cases}$$

Let $b_n = \frac{a_n}{n}$ for $n \in \mathbb{N}$. Then

- (A) both (a_n) and (b_n) are convergent
- (B) (a_n) is convergent but (b_n) is NOT convergent
- (C) (a_n) is NOT convergent but (b_n) is convergent
- (D) both (a_n) and (b_n) are NOT convergent

Question 39 (JAM 2023 • Q9 • ID: JAM_2023_Q9)

MCQ

+1 M

Neg: -0.33

Q. 9 Let $a_n = \left(1 + \frac{1}{n}\right)^n$ and $b_n = n \cos\left(\frac{n!\pi}{2^{10}}\right)$ for $n \in \mathbb{N}$. Then

- (A) (a_n) is convergent and (b_n) is bounded
- (B) (a_n) is NOT convergent and (b_n) is bounded
- (C) (a_n) is convergent and (b_n) is unbounded
- (D) (a_n) is NOT convergent and (b_n) is unbounded

Question 40 (JAM 2022 • Q55 • ID: JAM_2022_Q55)

NAT

+2 M

Neg: Nil (0)

Q.55

Let r be the radius of convergence of the power series

$$\frac{1}{3} + \frac{x}{5} + \frac{x^2}{3^2} + \frac{x^3}{5^2} + \frac{x^4}{3^3} + \frac{x^5}{5^3} + \frac{x^6}{3^4} + \frac{x^7}{5^4} + \dots$$

Then the value of r^2 is equal to _____. (Rounded off to two decimal places)

Question 41 (JAM 2022 • Q44 • ID: JAM_2022_Q44)

NAT

+1 M

Neg: Nil (0)

Q.44

The sum of the series

$$\sum_{n=1}^{\infty} \frac{1}{(4n-3)(4n+1)}$$

is equal to _____. (Rounded off to two decimal places)

Question 42 (JAM 2022 • Q38 • ID: JAM_2022_Q38)

MSQ

+2 M

Neg: Nil (0)

Q.38	Let (x_n) be a sequence of real numbers. Consider the set $P = \{n \in \mathbb{N} : x_n > x_m \text{ for all } m \in \mathbb{N} \text{ with } m > n\}.$ Then which of the following is/are true?
(A)	If P is finite, then (x_n) has a monotonically increasing subsequence.
(B)	If P is finite, then no subsequence of (x_n) is monotonically increasing.
(C)	If P is infinite, then (x_n) has a monotonically decreasing subsequence.
(D)	If P is infinite, then no subsequence of (x_n) is monotonically decreasing.

Question 43 (JAM 2022 • Q28 • ID: JAM_2022_Q28)

MCQ

+2 M

Neg: -0.67

Q.28	For $-1 \leq x \leq 1$, if $f(x)$ is the sum of the convergent power series $x + \frac{x^2}{2^2} + \frac{x^3}{3^2} + \cdots + \frac{x^n}{n^2} + \cdots$ then $f\left(\frac{1}{2}\right)$ is equal to
(A)	$\int_0^{\frac{1}{2}} \frac{\ln(1-t)}{t} dt.$
(B)	$-\int_0^{\frac{1}{2}} \frac{\ln(1-t)}{t} dt.$
(C)	$\int_0^{\frac{1}{2}} t \ln(1+t) dt.$
(D)	$\int_0^{\frac{1}{2}} t \ln(1-t) dt.$

Question 44 (JAM 2022 • Q14 • ID: JAM_2022_Q14)

MCQ

+2 M

Neg: -0.67

Q.14	The value of $\lim_{n \rightarrow \infty} \sum_{k=2}^n \frac{\sqrt{n+1} - \sqrt{n}}{k(\ln k)^2}$ is equal to
(A)	∞
(B)	1
(C)	e
(D)	0

Question 45 (JAM 2022 • Q12 • ID: JAM_2022_Q12)

MCQ

+2 M

Neg: -0.67

Q.12

Let T denote the sum of the convergent series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots + \frac{(-1)^{n+1}}{n} + \cdots$$

and let S denote the sum of the convergent series

$$1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \frac{1}{12} + \cdots = \sum_{n=1}^{\infty} a_n,$$

where

$$a_{3m-2} = \frac{1}{2m-1}, \quad a_{3m-1} = \frac{-1}{4m-2} \quad \text{and} \quad a_{3m} = \frac{-1}{4m} \quad \text{for } m \in \mathbb{N}.$$

Then which one of the following is true?

(A) $T = S$ and $S \neq 0$.(B) $2T = S$ and $S \neq 0$.(C) $T = 2S$ and $S \neq 0$.(D) $T = S = 0$.

Question 46 (JAM 2022 • Q8 • ID: JAM_2022_Q8)

MCQ

+1 M

Neg: -0.33

Q.8

Consider the series

$$\sum_{n=1}^{\infty} \frac{1}{n^m \left(1 + \frac{1}{n^p}\right)}$$

where m and p are real numbers.

Under which of the following conditions does the above series converge?

(A)

 $m > 1$.

(B)

 $0 < m < 1$ and $p > 1$.

(C)

 $0 < m \leq 1$ and $0 \leq p \leq 1$.

(D)

 $m = 1$ and $p > 1$.

Question 47 (JAM 2022 • Q7 • ID: JAM_2022_Q7)

MCQ

+1 M

Neg: -0.33

Q.7	Suppose $a_n = \frac{3^n + 3}{5^n - 5} \quad \text{and} \quad b_n = \frac{1}{(1 + n^2)^{\frac{1}{4}}} \quad \text{for } n = 2, 3, 4, \dots$ Then which one of the following is true?
(A)	Both $\sum_{n=2}^{\infty} a_n$ and $\sum_{n=2}^{\infty} b_n$ are convergent.
(B)	Both $\sum_{n=2}^{\infty} a_n$ and $\sum_{n=2}^{\infty} b_n$ are divergent.
(C)	$\sum_{n=2}^{\infty} a_n$ is convergent and $\sum_{n=2}^{\infty} b_n$ is divergent.
(D)	$\sum_{n=2}^{\infty} a_n$ is divergent and $\sum_{n=2}^{\infty} b_n$ is convergent.

Question 48 (JAM 2022 • Q6 • ID: JAM_2022_Q6)

MCQ

+1 M

Neg: -0.33

Q.6	Let (x_n) and (y_n) be sequences of real numbers defined by $x_1 = 1, \quad y_1 = \frac{1}{2}, \quad x_{n+1} = \frac{x_n + y_n}{2}, \quad \text{and} \quad y_{n+1} = \sqrt{x_n y_n} \quad \text{for all } n \in \mathbb{N}.$ Then which one of the following is true?
(A)	(x_n) is convergent, but (y_n) is not convergent.
(B)	(x_n) is not convergent, but (y_n) is convergent.
(C)	Both (x_n) and (y_n) are convergent and $\lim_{n \rightarrow \infty} x_n > \lim_{n \rightarrow \infty} y_n$.
(D)	Both (x_n) and (y_n) are convergent and $\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} y_n$.

Question 49 (JAM 2022 • Q5 • ID: JAM_2022_Q5)

MCQ

+1 M

Neg: -0.33

Q.5	The radius of convergence of the power series $\sum_{n=1}^{\infty} \left(\frac{n^3}{4^n} \right) x^{5n}$ is
(A)	4
(B)	$\sqrt[5]{4}$
(C)	$\frac{1}{4}$
(D)	$\frac{1}{\sqrt[5]{4}}$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Sequences and Series of Real Numbers (2022–2026) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key
1	NAT	+1	1.0 to 1.0	18	MCQ	+2	D	35	MCQ	+2	C
2	NAT	+1	1.0 to 1.0	19	MCQ	+1	C	36	MCQ	+2	B
3	NAT	+1	4.0 to 4.0	20	NAT	+2	0.98 to 1.02	37	MCQ	+2	B
4	MSQ	+2	C;D	21	NAT	+2	0.30 to 0.40	38	MCQ	+1	C
5	MCQ	+2	A	22	NAT	+1	1.95 to 2.05	39	MCQ	+1	C
6	MCQ	+2	C	23	MSQ	+2	B;C	40	NAT	+2	3.00 to 3.00
7	MCQ	+2	C	24	MSQ	+2	A;B	41	NAT	+1	0.24 to 0.26
8	MCQ	+1	D	25	MSQ	+2	A;B	42	MSQ	+2	A,C
9	MCQ	+1	D	26	MCQ	+2	B	43	MCQ	+2	B
10	NAT	+2	5.20 to 5.24	27	MCQ	+2	A	44	MCQ	+2	D
11	NAT	+1	1.98 to 2.02	28	MCQ	+2	A	45	MCQ	+2	C
12	MSQ	+2	A;C;D	29	MCQ	+2	D	46	MCQ	+1	A
13	MCQ	+2	B	30	NAT	+1	0.99 to 1.01	47	MCQ	+1	C
14	MCQ	+2	A	31	NAT	+1	0.49 to 0.51	48	MCQ	+1	D
15	MCQ	+2	B	32	MSQ	+2	A, D	49	MCQ	+1	B
16	MCQ	+2	A	33	MCQ	+2	B				
17	MCQ	+2	D	34	MCQ	+2	B				

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.